

Can Cross-Country Deep Learning Measure Subnational Development from Nightlights in the Balkans?

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Introduction

National GDP figures in the Western Balkans reveal little about where growth is occurring within countries. Subregional economic data is sparse and inconsistent, especially in poorer or more peripheral areas, even as the region contains both growing centers and places experiencing long-run depopulation. Satellite nightlight data offers a consistent alternative: VIIRS brightness captures economic activity at fine spatial scale, has been validated as a proxy for GDP [Hu and Yao, 2019], and can support cross-country prediction with convolutional neural networks [Yeh et al., 2020]. We ask whether those patterns transfer across the Balkans, whether a model trained on five countries can predict relative growth in a held-out sixth.

Data

The dataset covers six Balkan countries from 2013 to 2017: Albania, Bosnia and Herzegovina, Kosovo, Montenegro, North Macedonia, and Serbia. Using Google Earth Engine, we assemble a common 500 m raster for the full region. Model inputs include VIIRS 2013 baseline nightlight, built-up fraction, population density, elevation, slope, infrastructure proximity, and GHSL settlement class (SMOD). This means the model predicts later relative nightlight growth using only baseline nightlights and static covariates.

Each country is then extracted with its administrative boundary and divided into 32×32 pixel patches, or roughly 16 km tiles, with a stride of 16. This yields 5,353 patches across the six countries. Border patches are included, with out-of-country pixels zeroed. Each patch is labeled Low, Mid, or High using within-country terciles of mean log-ratio nightlight change from 2013 to 2017, which keeps class balance roughly even within each country.

Band	Channel	Source	Role
0	VIIRS 2013 baseline	VIIRS	input
4	Built-up fraction	GHSL	input
5	Population (log1p)	GHSL 2015	input
6	Elevation	SRTM	input
7	Slope	SRTM-derived	input
8	Infra proximity (log1p)	derived	input
9	SMOD code	GHSL SMOD	input
10	VIIRS 2017 endpoint	VIIRS	target construction

Table 1: Input channels and target source. All rasters were harmonized to a common 500 m grid. The model uses VIIRS 2013 plus static covariates as inputs, while VIIRS 2017 is used only to construct the 2013–2017 log-ratio target.

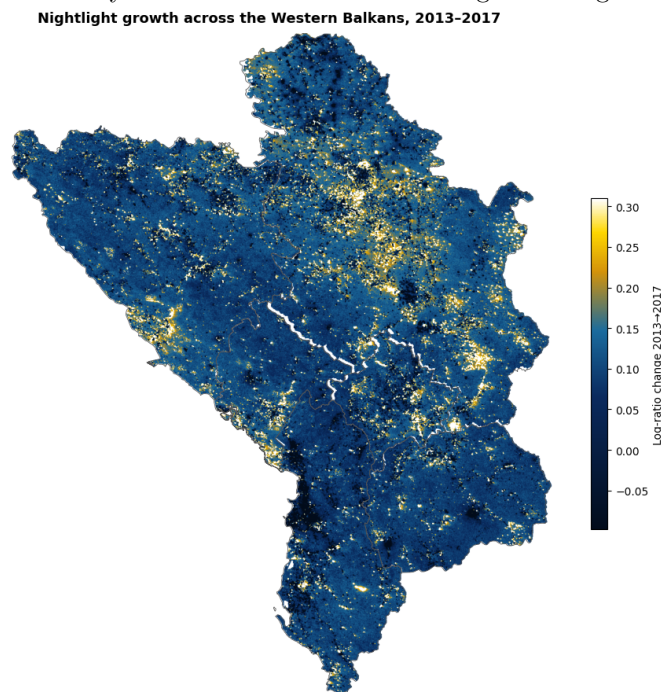


Figure 1: Log-ratio nightlight change across the Western Balkans, 2013–2017. Gold and white pixels indicate the strongest relative growth; dark blue indicates decline.

Methodology

We predict within-country terciles of log-ratio nightlight change from 2013 to 2017, using static geographic covariates and 2013 baseline VIIRS nightlight as inputs.

This matches the transfer question by testing whether relatively high-growth areas learned in five Balkan countries generalize to a held-out sixth, while preserving class balance. The log-ratio emphasizes proportional change, making growth more comparable across dim rural and already-bright urban areas.

A four-block CNN takes 7 input channels and maps each 32×32 patch to one of three classes: Low, Mid, or High change. Training uses AdamW, cosine annealing, early stopping, and random flips and rotations for data augmentation. Out-of-country pixels within each patch are zeroed, so the model attends only to in-country signal even for border patches.

Evaluation uses leave-one-country-out cross-validation: in each fold, the model trains on five countries and is tested on the sixth. Input channels are standardized within each fold using per-channel means and standard deviations computed from the training countries only, preventing leakage from the held-out country. A logistic regression on patch-mean features provides a non-spatial baseline.

Results

The 32×32 CNN achieves a mean accuracy of 40.3% and mean AUC of 0.582 across the six leave-one-country-out folds, above the 33% random baseline. Cross-country transfer is real but modest: the logistic regression baseline reaches 39.6% accuracy and 0.591 AUC, so the CNN’s margin over a non-spatial baseline is small (+0.7 pp accuracy). Bosnia and Herzegovina is the strongest fold (45.1%), while Albania and Serbia are the hardest. Kosovo and Serbia are the two folds where LR outperforms the CNN outright.

Country	N	Acc.	AUC
Albania	740	0.378	0.551
Bosnia & Herz.	1314	0.451	0.631
Kosovo	311	0.392	0.598
Montenegro	379	0.383	0.563
N. Macedonia	615	0.433	0.594
Serbia	1994	0.379	0.557
Mean	5353	0.403	0.582

Table 2: LOOCV results — log-ratio target, 7-channel input.

Using the GHSL settlement classification (SMOD) as an input also motivated a post hoc analysis of where transfer works best. A simple dominant-SMOD label at the patch level proved too coarse, because most 32×32 patches contain a mix of settlement types and therefore often appear overwhelmingly rural when reduced to a single class. Instead, we measured each patch’s *urban share*, defined as the fraction of pixels in urban SMOD categories, and compared performance across within-country terciles of that measure. We used within-country terciles rather than fixed regional thresholds because the six countries differ substantially in settlement structure: the “most urban” areas of Kosovo and Serbia are not directly comparable in absolute urban share, but they are comparable as the most urban portions of their own national landscapes.

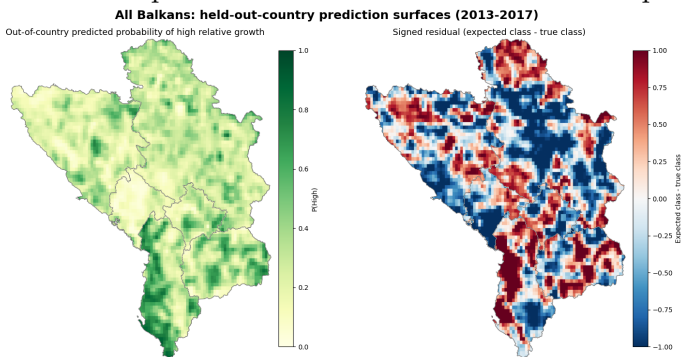


Figure 2: All-Balkans held-out-country prediction surfaces (2013–2017). Each country shown under the fold where it is the held-out test country.

This analysis suggests that out-of-country prediction is less stable in the relatively most urban patches within each country. The highest urban-share tercile has the largest mean absolute residual and the lowest accuracy, despite also having the highest average predicted probability of high growth, suggesting the model is more aggressive in dense built environments and more likely to overstate transferable high-growth signal.

Tercile	Resid.	AbsErr	Acc. (%)	$P(H)$
Low	-0.036	0.657	42.8	0.275
Mid	-0.078	0.698	41.2	0.281
High	-0.118	0.855	37.2	0.356

Table 3: Overall performance by within-country urban-share tercile. Higher absolute residual indicates worse transfer performance.

We interpret this cautiously. These terciles measure relative urbanization within each country rather than absolute urban classes, which was motivated by for our transfer setting: our question is not whether Kosovo and Albania contain identical urban shares, but whether the most urban parts of each country exhibit transferable growth patterns. Class composition still varies across bins, but the overall pattern suggests that these relatively urban areas are harder to transfer across countries, likely because urban growth processes are more heterogeneous than those in predominantly rural areas.

Conclusion

Cross-country transfer of nightlight growth patterns is real, even under the most demanding test: a model trained on five countries with no access to the sixth. The signal is not strong enough to replace ground-level data, but it is not empty either, and that is the more interesting finding. In a region where subnational economic data is sparse and inconsistently collected, the fact that structural conditions in one Balkan country say anything about growth trajectories in another is substantively meaningful.

The model’s modest performance suggests that transferable signal exists, but only weakly. Some broad patterns

in relative nightlight growth generalize across neighboring countries, yet patch-level prediction remains noisy, especially in more urban settings. This points to coarse shared structure in regional development rather than a strongly portable geography of growth.

The natural path forward is richer inputs. A single baseline snapshot, a coarse 500m grid, and six countries over four years is a minimal setup. Multispectral imagery, longer time horizons, and a broader set of training countries would each independently increase the signal available for transfer. The question posed here, whether neighbors can substitute for missing training data, remains open and worth pursuing at larger scale.

References

- Yingjie Hu and Yonglong Yao. Illuminating the dark side of economic development: Satellite nighttime light data and the measurement of poverty. *World Development*, 141: 105392, 2019.
- Christopher Yeh, Anthony Perez, Anne Driscoll, George Azari, Zhongyi Tang, David Lobell, Stefano Ermon, and Marshall Burke. Using publicly available satellite imagery and deep learning to understand economic well-being in africa. *Nature Communications*, 11(1):2583, 2020.